

(a) no sync: launches are queued

CPU  CPU runs ahead, queue stays full

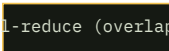


GPU stream 0



GPU stream 1

all-reduce (overlaps)



---

(b) `.item()` every step: the queue drains

CPU  `.item() waits` `.item() waits`

GPU stream 0



GPU idle

---

time →